Business Forecasting Practicals

Module 1 and Module 2: Aim, Steps and Conclusion

MODULE 1: Fundamentals of Forecasting and Time Series

1. Types of Forecasting: Qualitative and Quantitative

Aim

To understand qualitative and quantitative forecasting methods and to learn when each is used. Qualitative methods rely on opinion and suit cases with no past data, while quantitative methods use numerical data and models.

Steps

1. Define forecasting and note its use in production, inventory, budgeting and marketing.
2. List the qualitative methods: expert opinion, Delphi method, market survey and sales force opinion.
3. Note the features, advantages and limitations of each qualitative method.
4. List the quantitative methods: time series (moving average, trend) and causal (regression).
5. Note the data needed for each quantitative method.
6. Prepare a comparison table of the two types by data needed, cost, time and accuracy.
7. Take a case of a new product launch and choose a suitable method.
8. Take a case of a shop with five years of sales records and choose a suitable method.
9. Take a case where sales depend on advertising and choose a suitable method.
10. Write one reason for each choice and summarise the findings.

Conclusion

Qualitative methods are quick and flexible but can be biased. Quantitative methods are objective but need good past data, and businesses often combine both.

2. Data Collection, Classification and Presentation

Aim

To collect business data, classify it and present it in tables and charts for forecasting. Primary data is collected first-hand and secondary data comes from existing records and reports.

Steps

1. Decide the variable to study, for example monthly sales for 24 months.
2. Choose the source: a survey (primary) or company records and reports (secondary).
3. Collect the data and note the source.
4. Check for errors, missing values and repeated entries and correct them.
5. Classify the data by time, product or region.
6. Open Excel and enter the data in a table with clear headings.
7. Prepare a frequency table using `=COUNTIF()` or `=FREQUENCY()`.
8. Insert a bar chart to compare products.
9. Insert a line chart to show sales over time.
10. Add titles and axis labels and write two lines of observation.

Conclusion

Properly classified and presented data is easy to understand and gives a reliable base for forecasting. A forecast cannot be better than the data used to make it.

3. Introduction to Excel / Statistical Software for Forecasting

Aim

To become familiar with the features of Excel used for data entry, calculation, charting and forecasting. Excel has built-in functions such as AVERAGE, SLOPE, CORREL and FORECAST and the Data Analysis ToolPak.

Steps

1. Open Excel and identify the ribbon, formula bar, name box, cells, rows and columns.
2. Enter ten months of sales data in columns A and B.
3. Use `=SUM()`, `=AVERAGE()`, `=MAX()` and `=MIN()` on the sales column.
4. Use `=COUNT()` to count the entries.
5. Practise relative (B2) and absolute ($B$2) cell references.
6. Copy a formula down using the fill handle.
7. Enable the ToolPak through File, Options, Add-ins, Analysis ToolPak.
8. Run Descriptive Statistics from the Data tab.
9. Insert a line chart and a column chart from the Insert tab.
10. Save the file and take a printout.

Conclusion

Excel makes calculations and charts quick and accurate, and formulas update automatically when data changes. It is a useful base for all the forecasting practicals.

4. Components of Time Series

Aim

To identify the four components of a time series: trend, seasonal variation, cyclical variation and irregular variation. Trend is long-term movement, seasonal variation repeats within a year, cyclical variation lasts several years and irregular variation is unpredictable.

Steps

1. Collect monthly or quarterly sales data for 5 to 8 years.
2. Enter periods in column A and sales in column B.
3. Select the data and insert a line chart.
4. Add a linear trendline to see the long-term trend.
5. Note whether the trend is rising, falling or flat.
6. Look for patterns repeating each year and mark them as seasonal variation.
7. Look for longer waves of several years and mark them as cyclical variation.
8. Mark sudden unusual changes as irregular variation.
9. Write a short note on how each component appears in the data.
10. Write how each component could affect a business decision.

Conclusion

A time series combines trend, seasonal, cyclical and irregular movements. Identifying them helps in choosing the right forecasting method.

5. Moving Averages: Simple and Weighted

Aim

To smooth sales data and forecast using simple and weighted moving averages. The simple average gives equal weight to each period, while the weighted average gives more weight to recent periods and its weights add up to 1.

Steps

1. Enter months in column A and actual sales in column B.
2. Decide the period of the average, for example 3 months.
3. In C4 type `=AVERAGE(B2:B4)` for the simple moving average.
4. Drag the formula down to the last row.
5. Enter weights 0.2, 0.3 and 0.5 in cells F1 to F3.
6. In D4 type `=SUMPRODUCT(B2:B4,$F$1:$F$3)` and drag it down.
7. Take the last value of each column as the forecast for the next period.
8. Calculate the error and MAD for both methods.
9. Compare the MAD values.
10. Plot actual sales and both averages on one line chart.

Conclusion

Moving averages remove short-term fluctuations, and the weighted average reacts faster to recent changes. The method with the lower MAD is more accurate.

6. Trend Analysis: Linear Trend Method

Aim

To find the long-term trend using the least squares method and to forecast with the equation $Y = a + bX$. For example, with $X = 1$ to 5 and $Y = 10, 12, 15, 17, 21$, we get $b = 2.7$, $a = 6.9$ and the forecast for $X = 6$ is 23.1.

Steps

1. Enter the years in column A and sales (Y) in column B.
2. Code the years as X = 1, 2, 3 and so on in column C.
3. Add columns for XY and X^2.
4. Find ΣX, ΣY, ΣXY and ΣX^2 using `=SUM()`.
5. Calculate $b = (n\Sigma XY - \Sigma X\Sigma Y) / (n\Sigma X^2 - (\Sigma X)^2)$.
6. Calculate $a = (\Sigma Y - b\Sigma X) / n$.
7. Verify with `=SLOPE()` and `=INTERCEPT()`.
8. Write the trend equation $Y = a + bX$.
9. Forecast future periods by substituting the next X values.
10. Plot actual sales and the trend line on a chart.

Conclusion

The trend line shows the direction and rate of change of sales. It suits data with a steady rise or fall but does not capture seasonal changes.

7. Seasonal Variation: Simple Method

Aim

To measure seasonal variation using the simple average method. Each season is averaged over the years and compared with the grand average. An index above 100 shows a peak season and below 100 shows a slack season.

Steps

1. Collect quarterly or monthly data for 3 to 4 years.
2. Arrange it with years in rows and quarters in columns.
3. Find the total for each quarter.
4. Find the average of each quarter using `=AVERAGE()`.
5. Find the grand average of the quarterly averages.
6. Calculate the seasonal index = (Quarter average ÷ Grand average) × 100.
7. Check that the indices total 400 (or 1200 for months).
8. Draw a column chart of the seasonal indices.
9. Identify the peak and slack seasons.
10. Adjust a forecast using: Seasonal forecast = Average forecast × (Index ÷ 100).

Conclusion

Seasonal indices show strong and weak seasons, which helps in planning stock, production and staffing. The method works best when there is no strong trend.

8. Practical Problems Using Spreadsheet / Software

Aim

To solve forecasting problems in Excel and choose the best method by comparing errors such as MAD, MSE and MAPE. The method with the smallest error is the best fit.

Steps

1. Take the given time series problem, for example 12 months of sales.
2. Enter the data in Excel.
3. Draw a line chart to study the pattern.
4. Apply the 3-period simple and weighted moving averages.
5. Fit the linear trend using `=SLOPE()` and `=INTERCEPT()`.
6. Calculate seasonal indices if the data is seasonal.
7. Calculate the error as Actual minus Forecast for each method.
8. Find MAD, MSE and MAPE for each method.
9. Select the method with the lowest error.
10. Plot the final chart and write the forecast for the next period.

Conclusion

Spreadsheets solve forecasting problems quickly and accurately. Comparing errors helps to choose the best method, and forecasts should be updated when new data arrives.

MODULE 2: Regression and Business Forecasting Applications

1. Concept of Correlation

Aim

To understand correlation between two variables using a scatter diagram. Correlation is positive when both variables move together, negative when they move in opposite directions and zero when there is no pattern.

Steps

1. Take two related variables, such as advertising and sales, for 12 months.
2. Enter X in column A and Y in column B.
3. Select both columns.
4. Insert a Scatter (XY) chart from the Insert tab.
5. Add the chart title and axis labels.
6. Observe whether the points rise, fall or are scattered.
7. Note how closely the points lie around a straight line.
8. State whether the correlation is positive, negative or none.
9. Repeat with another pair, such as price and demand.
10. Compare the two diagrams and write your observations.

Conclusion

Correlation shows the direction and strength of a relationship between two variables. It does not prove cause and effect.

2. Karl Pearson Correlation Coefficient

Aim

To calculate Karl Pearson's coefficient of correlation (r), which lies between -1 and $+1$. A value near $+1$ means strong positive correlation, near -1 means strong negative correlation and near 0 means no linear correlation.

Steps

1. Enter X in column A and Y in column B.
2. Add helper columns for XY, X^2 and Y^2.
3. Find n using `=COUNT()`.
4. Find ΣX, ΣY, ΣXY, ΣX^2 and ΣY^2 using `=SUM()`.
5. Calculate the numerator $n\Sigma XY - \Sigma X\Sigma Y$.
6. Calculate the denominator $\sqrt{[(n\Sigma X^2 - (\Sigma X)^2)(n\Sigma Y^2 - (\Sigma Y)^2)]}$.
7. Divide to get r.
8. Check the result with `=CORREL()`. For X = 2, 4, 6, 8, 10 and Y = 5, 9, 12, 16, 18, r = 0.995.
9. Calculate r^2 to find the proportion of variation explained.
10. Interpret the value of r.

Conclusion

The value of r showed a strong positive relationship, and the manual result matched the CORREL function. The coefficient measures only linear relationships.

3. Linear Regression Method

Aim

To build a regression equation $Y = a + bX$, where Y is the dependent and X the independent variable, and to measure the fit using R^2. For the data in the previous practical, $Y = 2.1 + 1.65X$.

Steps

1. Enter X (independent) in column A and Y (dependent) in column B.
2. Find ΣX, ΣY, ΣXY and ΣX^2.
3. Calculate $b = (n\Sigma XY - \Sigma X\Sigma Y) / (n\Sigma X^2 - (\Sigma X)^2)$.
4. Calculate $a = (\Sigma Y - b\Sigma X) / n$.
5. Verify with `=SLOPE()` and `=INTERCEPT()`.
6. Write the equation $Y = a + bX$.
7. Find R^2 using `=RSQ()`.
8. Insert a scatter chart of X and Y.
9. Add a linear trendline showing the equation and R^2.
10. Interpret the slope, intercept and R^2 in simple words.

Conclusion

The equation shows how Y changes when X changes by one unit. A high R^2 means the line fits the data well.

4. Forecasting Using Regression Equation

Aim

To forecast values using a fitted regression equation by substituting new values of X. Forecasts are most reliable within the range of the original data. For example, with $Y = 2.1 + 1.65X$, the forecast for $X = 12$ is 21.9.

Steps

1. Enter the past data of X and Y.
2. Find the slope using `=SLOPE()`.
3. Find the intercept using `=INTERCEPT()`.
4. Write the regression equation $Y = a + bX$.
5. Enter the new X values in a separate column.
6. Calculate the forecast by the formula `=intercept+slope*X`.
7. Check the results with `=FORECAST(x, known_y, known_x)`.
8. Compare with actual values when available and find the error.
9. Calculate MAD.
10. Plot the actual data, the regression line and the forecast points.

Conclusion

Regression gives useful forecasts for planned values of X. Accuracy falls if conditions change, so the model should be updated regularly.

5. Demand Forecasting

Aim

To forecast the future demand for a product using past demand and factors such as price or income. Demand forecasts help in production planning, inventory control and purchasing.

Steps

1. Collect 12 to 24 months of demand data.
2. Collect the related factor, such as price or income.
3. Enter the data in Excel with columns for period, demand and factor.
4. Plot demand over time.
5. Decide whether trend or seasonality is present.
6. Fit a linear trend or a regression of demand on price.
7. Forecast the demand for the next period.
8. Adjust the forecast using the seasonal index if needed.
9. Plot actual and forecast demand.
10. State the number of units to produce or stock.

Conclusion

Demand forecasting helps to avoid overstocking and shortages. Forecasts are never exact, so safety stock is kept and the model is reviewed regularly.

6. Sales Forecasting

Aim

To forecast the future sales of a business using moving average, trend or regression methods. A sales forecast is the base for the sales budget, production budget and marketing plans.

Steps

1. Enter monthly or yearly sales data in Excel.
2. Draw a line chart to study the trend and seasonal pattern.
3. Apply the 3-period moving average.
4. Fit a linear trend and forecast the next periods.
5. If advertising data is available, fit a regression of sales on advertising.
6. Calculate the forecast error for each method.
7. Calculate MAD for each method.
8. Select the most accurate method.
9. Convert the forecast into a sales target for each month or region.
10. Prepare a chart and table of actual and forecast sales.

Conclusion

Sales forecasts help to set targets, budgets and marketing plans. They should be revised when prices, competition or customer taste change.

7. Index Numbers: Basics and Applications

Aim

To calculate index numbers and use them in business. An index number shows the change in prices or quantities compared with a base period taken as 100. Laspeyres uses base-year quantities, Paasche uses current-year quantities and Fisher is the geometric mean of the two.

Steps

1. Enter the items with base-year price (P0) and current-year price (P1).
2. Enter the base-year quantity (Q0) and current-year quantity (Q1).
3. Calculate the simple price index = (P1 ÷ P0) × 100 for each item.
4. Add columns for P1Q0, P0Q0, P1Q1 and P0Q1.
5. Find the totals using `=SUM()`.
6. Laspeyres = (ΣP1Q0 ÷ ΣP0Q0) × 100.
7. Paasche = (ΣP1Q1 ÷ ΣP0Q1) × 100.
8. Fisher = `=SQRT(Laspeyres*Paasche)`.
9. Interpret the result. An index of 114 means prices are 14 percent higher than the base year.
10. Deflate sales: Real sales = (Current sales ÷ Price index) × 100.

Conclusion

Index numbers give a single figure for the change in prices over time. They are used to measure inflation, adjust wages and deflate sales figures.

8. Forecasting Using Real Business Data

Aim

To apply forecasting methods to real company data, such as sales or profit from annual reports or stock exchange websites. Real data is usually less regular than textbook data.

Steps

1. Select a company and a variable, for example 10 years of net sales.
2. Download the data from a reliable source.
3. Note the source and the units.
4. Clean the data by correcting errors and filling gaps sensibly.
5. Enter the data in Excel.
6. Plot a line chart to identify trend and seasonality.
7. Apply the trend, moving average or regression method.
8. Keep the last few observations aside and forecast them.
9. Calculate MAD and MAPE to measure accuracy.
10. Forecast the next period and write a short interpretation.

Conclusion

Forecasting methods give useful estimates even for irregular real data when applied carefully. Results should be used along with knowledge of the company and the economy.

9. Case Study / Practical Assignments Using Software

Aim

To solve a business forecasting case study in Excel and give a recommendation. The case requires defining the problem, choosing a method, testing it and interpreting the results.

Steps

1. Read the case and note the problem and the objective.
2. List the data given.
3. Enter the data in Excel.
4. Draw charts to understand the pattern.
5. Choose a suitable method: time series or regression.
6. Do the calculations using formulas and functions.
7. Test the accuracy using MAD or R^2.
8. Prepare the forecast for the required period.
9. State the assumptions and limitations.
10. Write the final recommendation, such as the quantity to produce or the budget to approve.

Conclusion

Forecasting tools help managers make informed decisions on production, sales and budgeting. Forecasts are estimates and should be reviewed regularly.